

FLOATING HIGH

HOT AIR BALLOONS

Unlike speedy planes, hot air balloons float across the sky slowly. French brothers Joseph-Michel and Jacques-Étienne Montgolfier carried out the first balloon flight in 1793 with a balloon made of silk and paper. Hot air balloons today are made of materials such as nylon or polyester, with a flameproof skirt at the bottom for safety. They are filled with hot air using a burner. As the air warms up, the particles move faster, spread out, and take up more space, making the air less dense than the cooler air outside the balloon. This generates a force called upthrust, which causes the balloon to rise up in the sky.

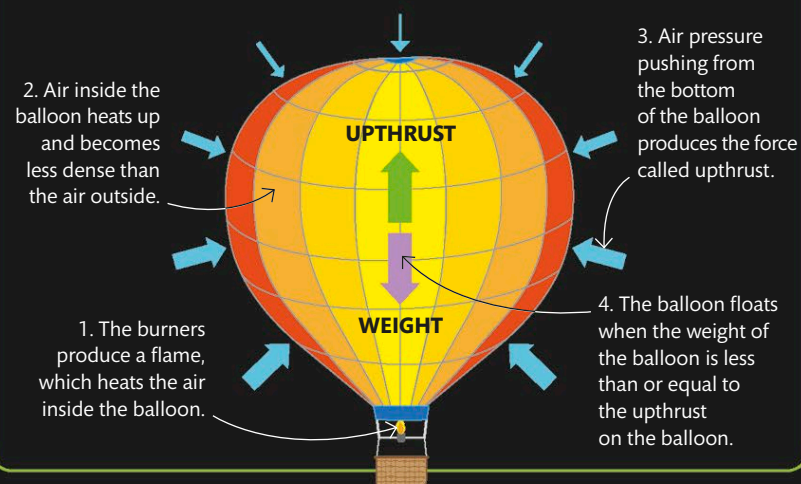
FLYING AIRSHIPS

Airships, like this one in Germany, are filled with helium gas. Just like hot air, helium is less dense than the air in which it floats. Airships have propellers and rudders, which direct them through the air, while a hot air balloon flies in whichever direction the wind blows.



UPTHRUST

An upward force, called upthrust, pushes a hot air balloon upward. Upthrust is produced by air pressure, which pushes in all directions but is stronger below the balloon than above it.



The world's largest passenger hot air balloon is **131 ft (40 m)** tall and can carry **32 people** at a time.